

Oracle SQL Subquery

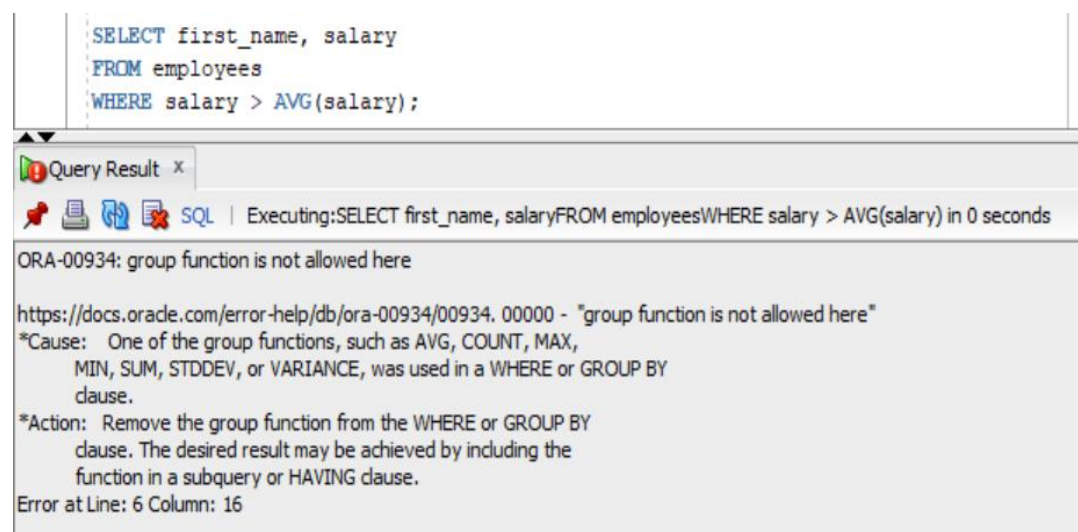
Query Review 1

Original Query:

```
SELECT first_name, salary FROM employees WHERE salary >
AVG(salary);
```

Error:

-ORA-00934: group function is not allowed here



Explanation:

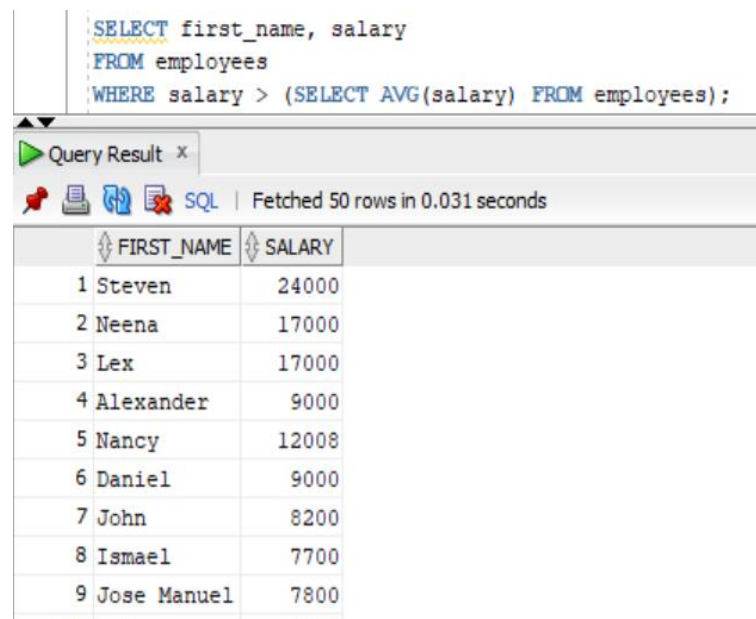
-AVG is an aggregate function and cannot be used directly in the WHERE clause.

-Oracle must calculate the average salary first using a subquery.

-Then the outer query compares each employee salary with that value.

Correct Query:

SELECT first_name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);



The screenshot shows a SQL query execution window. The query is: `SELECT first_name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);`. The results are displayed in a table with two columns: FIRST_NAME and SALARY. The table contains 9 rows of data, sorted by salary in descending order. The first row is Steven with a salary of 24000, and the last row is Jose Manuel with a salary of 7800. The interface also shows a status bar indicating 'Fetches 50 rows in 0.031 seconds'.

	FIRST_NAME	SALARY
1	Steven	24000
2	Neena	17000
3	Lex	17000
4	Alexander	9000
5	Nancy	12008
6	Daniel	9000
7	John	8200
8	Ismael	7700
9	Jose Manuel	7800

- **Type:** Scalar subquery

- **Business Logic:** Returns employees whose salary is higher than the company average salary.

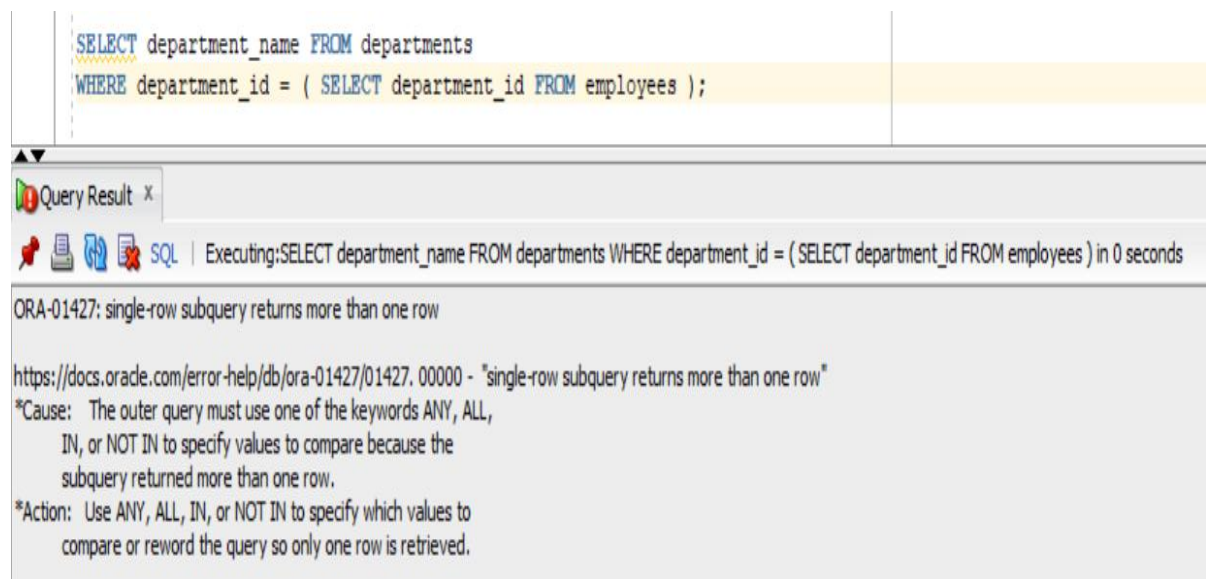
Query Review 2

Original Query:

```
SELECT department_name FROM departments WHERE  
department_id = ( SELECT department_id FROM employees );
```

Error:

ORA-01427: single-row subquery returns more than one row



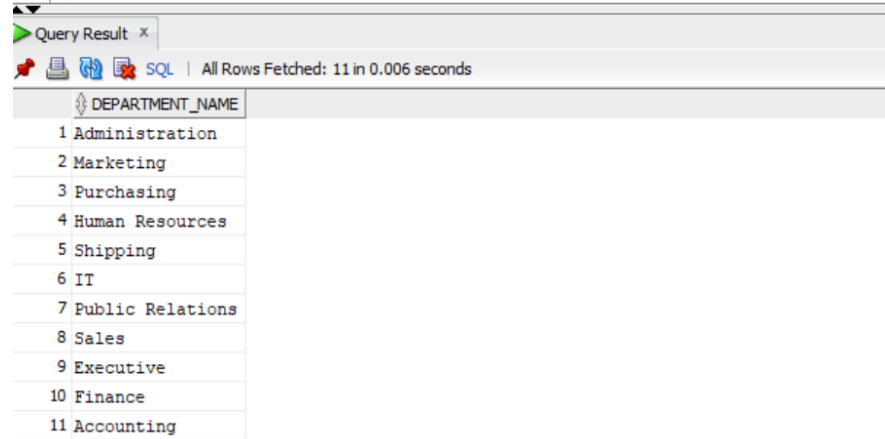
Explanation:

- The subquery returns many department_id values from the employees table.
- The = operator can compare with one value only, so Oracle gives an error.

- Correct Query:

SELECT department_name FROM departments WHERE
department_id IN (SELECT DISTINCT department_id FROM
employees WHERE department_id IS NOT NULL);

```
SELECT department_name FROM departments WHERE department_id IN  
( SELECT DISTINCT department_id FROM employees WHERE department_id IS NOT NULL);
```



DEPARTMENT_NAME
1 Administration
2 Marketing
3 Purchasing
4 Human Resources
5 Shipping
6 IT
7 Public Relations
8 Sales
9 Executive
10 Finance
11 Accounting

- Subquery Type: Multi-row Subquery

- Business Logic: Returns departments that currently have employees assigned to them.

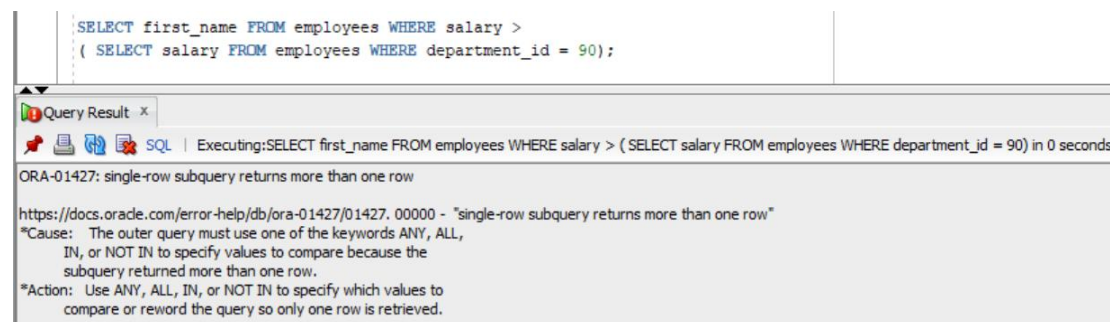
Query Review 4

Original Query:

```
SELECT first_name FROM employees WHERE salary > ( SELECT  
salary FROM employees WHERE department_id = 90);
```

Error:

ORA-01427: single-row subquery returns more than one row

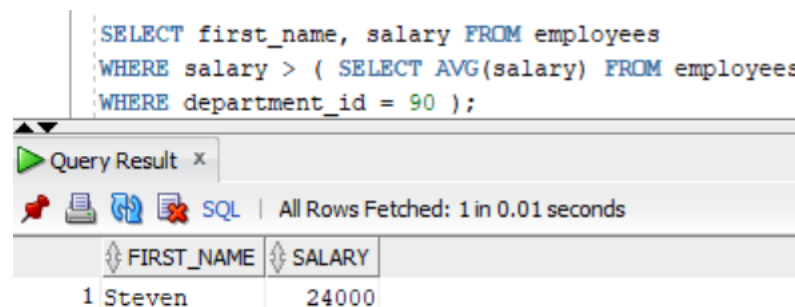


Explanation:

- The subquery returns multiple salary values from department 90.
- The > operator requires a single value for comparison.
- Therefore Oracle raises an error.

Correct Query:

```
SELECT first_name, salary FROM employees  
WHERE salary > ( SELECT AVG(salary) FROM employees  
WHERE department_id = 90 );
```



Subquery Type: Scalar Subquery

Business Logic: Returns employees whose salary is greater than the average salary of employees in department 90.